



Data-Driven Decision-Making

Lorem Ipsum Hotel Group: Hotel Cancellation Prediction Challenge

Group 72

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1. Executive Summary

The surge in reservation cancellations at the Lorem Ipsum Hotel Group has impacted occupancy rates, revenue generation, and customer satisfaction. Previous counterstrategies have been proven ineffective, resulting in costs with guest relocations and long-term reputational damage. To reverse this trend, this project developed a predictive analytics solution designed to anticipate the cancellation probability of individual bookings well before the guest's scheduled arrival.

By leveraging historical reservation data within the KNIME Analytics Platform, the data pipeline will resolve data inconsistencies and engineer critical behavioural metrics (like total length of stay and a custom, history-based cancellation ratio). Multiple machine learning classifiers (Decision Trees and Random Forests) will be developed and evaluated, as the financial risk of failing to catch a true cancellation is high. The model selection process prioritizes Recall (Sensitivity) while maintaining a balance with Precision and F1-score.

Integrated into an automated workflow, the model generates a practical risk ledger for hotel management. Instead of relying on generalized overbooking tactics, the hotel will now apply targeted, data-driven interventions, such as personalized confirmation prompts, calibrated deposit requirements, or proactive rebooking campaigns. To preserve the system's long-term accuracy against market seasonality, a continuous performance monitoring framework and periodic data quality audits will be established for routine retraining.

This transition intend on empowering the hotel group to shift from a reactive approach to a proactive and predictive strategy.

2. Business and Data Understanding

To better understand the dataset, we started by analyzing the available documentation and using visualization nodes to identify the nature and distribution of each variable. There are some variables that are clearly categorical and for that we applied ruled based mapping to convert the string variables into integer categories to facilitate the model's performance. The variables transformed were: Meal, ReservedRoomType, DepositType, CustomerType, DistributionChannel, MarketSegment.

During the exploratory analysis using the Statistics node, some inconsistencies were identified and corrected. The Adult variable contained observations with a value of zero,

which we did not find to be realistic given that a reservation requires at least one adult. Therefore, these values were replaced with one.

The Children variable was considered a float variable, and since it should be a discrete variable, we applied the floor function so it would assume the lower integer number.

The analysis of the Babies variable through a boxplot showed that most reservations did not include babies, while reservations that did include babies presented a wider range of values. Based on this distribution, the variable was transformed into a categorical feature with three groups: no babies, one or two babies, and more than two babies.

Regarding feature engineering, two new variables were created. The first variable, LengthOfStays, represents the total duration of a customer's stay by adding the values of StaysInWeekendNights and StaysInWeekNights. This provides a more complete representation of the reservation duration. The second variable, CancellationRatio, was created by combining information about previous cancellations and previous non-cancelled bookings, representing the proportion of cancelled bookings relative to the customer's reservation history.

There were missing values identified in the Country and Children variables. For Country, the missing values were replaced with the most frequent category, as this represented the dominant value in the dataset. For Children, missing values could represent either the absence of children or a missing entry. And since most reservations contained zero children, it was clear that the best approach was to replace the missing values with zero.

Outlier analysis was performed using boxplots and statistical measures. Initially, several variables appeared to contain outliers, however, some were excluded by being categorical variables and variables with values that were logically explainable due to the nature of the data. That was the case of the Babies variable, that upon further analysis of its distribution, instead of treating the bigger values as outliers, we changed it into categories as explained before.

Only ADR and LengthOfStays presented relevant outliers, replacing them with acceptable values.

Although DaysInWaitingList presented a highly skewed distribution, the observed values were considered realistic. Two approaches were tested: maintaining the original

continuous variable and transforming it into intervals. The original format resulted in better model performance, so the continuous representation was kept.

Finally, a correlation matrix was used to analyse relationships between variables and identify redundant features. After creating the CancellationRatio variable, it was expected that PreviousCancellations and PreviousBookingNotCancelled would provide limited additional information. The correlation analysis confirmed that these variables were highly related, so they were removed to reduce redundancy. The same could be thought about StaysInWeekNights, StaysInWeekendNights and LengthOfStay, but upon further testing, removing any of these variables affected the model's performance so they were all kept.

The same analysis was applied to DistributionChannel and BookingID. Since these variables showed little correlation with the target variable, they were excluded from the final model because they were unlikely to contribute meaningful predictive information.

Since some variables had different scales, the dataset was normalized to ensure that features with larger numerical ranges did not disproportionately influence the model.

3. Model Engineering and Evaluation

The models trained and evaluated were a Decision Tree and three Random Forests with different numbers of estimators (100, 150, and 200).

These models were used to validate the hypotheses and insights obtained during the exploratory data analysis. For example, DistributionChannel presented a low correlation with the target variable, leading us to expect that it would have limited predictive value. This was confirmed during model evaluation, as removing this variable did not negatively affect performance. However, other variables, such as Meal, also presented low correlation with the target variable but improved model performance when included, demonstrating that correlation alone is not sufficient to determine a variable's importance.

The objective of the classification model is to identify reservations that are likely to be cancelled. From a business perspective, failing to identify a future cancellation represents a greater risk than incorrectly predicting a cancellation. A False Negative may result in unexpected revenue losses greater than the costs of trying to retain a client that

wasn't going to cancel, a False Positive. Therefore, the main priority was to minimize False Negatives by maximizing Recall / Sensitivity.

Although Recall was the primary evaluation metric, other metrics were also considered to ensure a balanced model performance. The F1-score was analysed to evaluate the trade-off between Recall and Precision, while the F2-score was also considered due to assigning a higher importance to Recall compared to F1-score.

Several model iterations were performed to evaluate which variables contributed most. The experiments showed that some variables affected different algorithms in different ways. For example, including the LengthOfStays variable decreased the performance of the Decision Tree model but improved the performance of the Random Forest models.

If we look strictly at catching every single cancellation to maximize our sensitivity, the standalone Decision Tree scored slightly higher at 0.778 compared to the Random Forest's 0.748. However, the 100-tree Random Forest gave us a much cleaner balance across the board, showing a jump in precision to 0.882 and a better overall F1-score of 0.809. Overall, the Random Forest models consistently outperformed the Decision Tree. When comparing the Random Forest models with more estimators, the performance metrics completely flattened out, keeping it at 100 trees would save us from wasting unnecessary computing power.

4. Deployment

To make this practical for day-to-day operations, we set up the automated pipeline in KNIME using a Random Forest Predictor node. Before the model can flag incoming reservations, the live data is automatically routed through the same preprocessing and cleanup steps we established during the training phase to ensure consistency. The pipeline corrects data anomalies by forcing the adult variable to one if it was left at zero, dropping children floats down to whole integers using a floor function, and grouping the baby variable into simple categories of zero, one or two, or more than two babies. In the same step, the system automatically builds our newly engineered metrics, combining weekend and weeknights into a total length of stay and calculating our custom cancellation ratio. Finally, it filters out redundant or useless data. As we created the cancellation ratio, old metrics (like previous cancellations and previous bookings not cancelled) become highly correlated and are safely dropped, alongside with irrelevant identifiers like the booking ID.

Once this data is fully processed, the model calculates the probability of each guest cancelling. The final predictions are exported into a CSV file, giving the managers a list of high-risk bookings. This allows the team to take preventive actions, whether that means

sending out a friendly confirmation message, securing a deposit, or adjusting booking management strategies.

5. Monitoring and Maintenance

Once the model is live, it requires continuous monitoring to ensure the predictions remain accurate over time. Since our main goal is to minimize missed cancellations and protect revenue, Recall will serve as our primary health indicator: a noticeable drop in Recall means the model is failing to catch cancellation-prone bookings, which will serve as our trigger to retrain the system. At the same time, we will track Precision, Accuracy, and F1-scores to ensure the model maintains a healthy operational balance and doesn't begin over-reporting false alarms.

We also need to run periodic data quality audits to check for missing values or unexpected data distributions. We will pay attention to the heavy hitters in our model, specifically key variables like Lead Time, Deposit Type, our engineered Length of Stay, and the custom Cancellation Ratio.

Furthermore, guest booking behavior naturally shifts over time due to seasonality, changing economic climates, or updates to the hotel's internal policies. When these shifts happen, the older training data will lose its relevance, so, by setting up a routine maintenance schedule to retrain the Random Forest model on fresh reservation data, we can ensure the pipeline adapts to current booking trends and continues to provide reliable insights.

6. Conclusion

This project demonstrates how predictive analytics can successfully transform revenue management at the Lorem Ipsum Hotel Group. By comparing historical reservation data, we moved away from failed and reactive tactics (like aggressive overbooking and punitive cancellation policies), and developed a proactive machine learning pipeline capable of catching high-risk cancellations before arrival.

A key driver of this success was our focus on data preprocessing and feature engineering. Resolving data anomalies and creating tailored metrics, such as LengthOfStays and CancellationRatio, was essential to improve data quality and uncover some customer behavior patterns.

Among the models evaluated, the 100-tree Random Forest delivered the optimal operational balance. By prioritizing Recall, we ensured that the hotel effectively flags cancellation-prone bookings while keeping false alarms to a minimum.

Furthermore, in production, the automated KNIME pipeline translates these technical predictions into direct business value. Rather than applying generalized rules, hotel managers can now use the model's risk output to deploy targeted interventions. This includes sending personalized confirmation prompts, adjusting deposit criteria for high-risk guests, or calibrating safe overbooking thresholds to maximize room occupancy.

However, deployment is a continuous cycle rather than a final milestone, as customer behaviour fluctuates with seasonality and shifting market trends. That said, the model requires routine data quality audits and periodic retraining to prevent performance drift.

Looking ahead, the hotel group could further refine its predictive power by testing advanced algorithms like Gradient Boosting or XGBoost, conducting deeper hyperparameter tuning, and integrating external data (like local event calendars or economic indicators).

Ultimately, this framework bridges the gap between raw data and business strategy, empowering the hotel group to secure its revenue while enhancing the overall guest experience.